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INTERFACE TECHNIQUES FOR STACKED MEMORY ARCHITECTURES

Abstract

Methods, systems, and devices for interface techniques for stacked memory architectures are described. A semiconductor system, such as a memory system, may distribute memory access circuitry among multiple semiconductor dies of a stack. A first die of the system may include logic circuitry operable to configure a set of multiple first interface blocks of the first die. Each first interface block may include circuitry operable to communicate with one or more second interface blocks of one or more second dies of the system to access a respective set of one or more memory arrays of the one or more second dies. In some examples, the system may include a respective controller for each first interface block to support access operations via the first interface block. The system may also include non-volatile storage, one or more sensors, or a combination thereof to support various operations of the system.

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Background/Summary

CROSS REFERENCE [0001] The present Application for Patent claims priority to U.S. Patent Application No. 63/470,676 by Akel et al., entitled “INTERFACE TECHNIQUES FOR STACKED MEMORY ARCHITECTURES,” filed Jun. 2, 2023, which is assigned to the assignee hereof, and which is expressly incorporated by reference in its entirety herein.

TECHNICAL FIELD
[0002] The following relates to one or more systems for memory, including interface techniques for stacked memory architectures.
BACKGROUND
[0003] Memory devices are widely used to store information in devices such as computers, user devices, wireless communication devices, cameras, digital displays, and others. Information is stored by programming memory cells within a memory device to various states. For example, binary memory cells may be programmed to one of two supported states, often denoted by a logic 1 or a logic 0. In some examples, a single memory cell may support more than two states, any one of which may be stored. To access the stored information, the memory device may read (e.g., sense, detect, retrieve, determine) states from the memory cells. To store information, the memory device may write (e.g., program, set, assign) states to the memory cells.
[0004] Various types of memory devices exist, including magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), static RAM (SRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), self-selecting memory, chalcogenide memory technologies, not-or (NOR) and not-and (NAND) memory devices, and others. Memory cells may be described in terms of volatile configurations or non-volatile configurations. Memory cells in a non-volatile configuration may maintain stored logic states for extended periods of time even in the absence of an external power source. Memory cells in a volatile configuration may lose stored states when disconnected from an external power source.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1** shows an example of a system that supports interface techniques for stacked memory

architectures in accordance with examples as disclosed herein.
[0006] FIG. **2** shows an example of a system that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein.
[0007] FIGS. **3** and **4** show examples of an interface architecture that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein.
[0008] FIG. **5** shows a block diagram of a memory system that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein.
[0009] FIG. **6** shows a block diagram of a system that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein.
[0010] FIGS. **7** and **8** show flowcharts illustrating methods that support interface techniques for stacked memory architectures in accordance with examples as disclosed herein.
DETAILED DESCRIPTION
[0011] Some memory systems may include a stack of semiconductor dies, including one or more memory dies above a logic die that is operable to access a set of memory arrays distributed across the one or more memory dies. Such an architecture may be implemented as part of a coupled dynamic random access memory (DRAM) system, and may support solutions for memory-centric logic, such as graphics processing units (GPUs), among other implementations. In some examples, a 3D stacked memory system may be closely coupled (e.g., physically coupled, electrically coupled) with a processor, such as a GPU or other host, as part of a physical memory map accessible to the processor. Such coupling may include one or more processors being implemented in a same semiconductor die as at least a portion of a 3D stacked memory system (e.g., as part of a logic die), or a processor being implemented in a die that is directly coupled (e.g., fused) with another die that includes at least a portion of a 3D stacked memory system. Unlike cache-based memory, 3D stacked memory system may not be backed by a level of external memory with the same physical addresses. For example, a 3D stacked memory system may be associated with and located within a dedicated base address, where each portion of the 3D stacked memory system may be non-overlapping within the address.
[0012] In accordance with examples as disclosed herein, a semiconductor system, such as a 3D stacked memory system or other memory system, may distribute memory access circuitry among multiple semiconductor dies of a stack (e.g., a stack of multiple directly-coupled semiconductor dies). For example, a first die (e.g., a logic die) of the system may include a logic block (e.g., a common logic block, a central logic block, logic circuitry) operable to configure a set of multiple first interface blocks (e.g., memory interface blocks, instances of first interface circuitry) of the first die. In some examples, the system may include a respective controller (e.g., a memory controller, a host interface controller, of the first die, of another die) for each first interface block to support access operations via the first interface block. The system may also include non-volatile storage, one or more sensors (e.g., temperature sensors, current sensors, voltage sensors, counters), or a combination thereof to support various operations of the system.
[0013] The set of first interface blocks may be configured to access one or more memory arrays of one or more second dies (e.g., array dies). For example, a first interface block of the first die may communicate with one or more second interface blocks (e.g., one or more instances of second interface circuitry) of a second die to access a respective set of one or more memory arrays (e.g., of the second die). In some examples, a first interface block (e.g., of the first die) may include circuitry to perform or otherwise support operations, such as initialization operations, evaluation operations, configuration operations, access operations, scheduling operations, repair operations, refresh operations, error control operations, adverse access operations, signaling operations, or other operations based on information (e.g., signaling, instructions, parameters, configuration information) of the system. However, various examples of the described techniques may include other distribution of memory access functionality among circuitry of multiple semiconductor dies.
[0014] A first interface block and a second interface block configured to access a corresponding set

of one or more memory arrays may be communicatively coupled between dies using one or more of various interconnection techniques, such as a fusion of conductive contacts of the respective dies, which may enable a relatively higher density of contacts than other techniques. In some examples, a first die may also include aspects of a host system (e.g., a host processor), which may further reduce limitations associated with memory die interconnections. Such an architecture may be extended by including multiple sets of memory arrays on a given second die, each with a respective second interface block, or by stacking a set of multiple second dies over a given first die, or both, such that a given second die may include a respective second portion of the access circuitry for each set of memory arrays of the given second die. By dividing memory access circuitry among multiple semiconductor dies in accordance with one or more of the described techniques, a system may be configured with an increased throughput of information, or a greater storage density, among other advantages, compared with other techniques for configuring a memory system.

[0015] Features of the disclosure are illustrated and described in the context of systems and dies. Features of the disclosure are further illustrated and described in the context of interface architectures and flowcharts.

[0016] FIG. **1** illustrates an example of a system **100** that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The system **100** may include portions of an electronic device, such as a computing device, a mobile computing device, a wireless communications device, a graphics processing device, a vehicle, or other systems. The system **100** includes a host system **105**, a memory system **110**, and one or more channels **115** coupling the host system **105** with the memory system **110** (e.g., to provide a communicative coupling). The system **100** may include one or more memory systems **110**, but aspects of the one or more memory systems **110** may be described in the context of a single memory system **110**.

[0017] The host system **105** may be an example of a processor (e.g., circuitry, processing circuitry, one or more processing components) that uses memory to execute processes, such as a processing system of a computing device, a mobile computing device, a wireless communications device, a graphics processing device, a wearable device, an internet-connected device, a vehicle controller, a system on a chip (SoC), or other stationary or portable electronic device, among other examples. The host system **105** may include one or more of an external memory controller **120**, a processor **125**, a basic input/output system (BIOS) component **130**, or other components (e.g., a peripheral component, an input/output controller, not shown). The components of the host system **105** may be coupled with one another using a bus **135**.

[0018] An external memory controller **120** may be configured to enable communication of information (e.g., data, commands, control information, configuration information) between components of the system **100** (e.g., between components of the host system **105**, such as the processor **125**, and the memory system **110**). An external memory controller **120** may process (e.g., convert, translate) communications exchanged between the host system **105** and the memory system **110**. In some examples, an external memory controller **120**, or other component of the system **100**, or associated functions described herein, may be implemented by or be part of the processor **125**. For example, an external memory controller **120** may be hardware, firmware, or software (e.g., instructions), or some combination thereof implemented by a processor **125** or other component of the system **100** or the host system **105**. Although an external memory controller **120** is illustrated outside the memory system **110**, in some examples, an external memory controller **120**, or its functions described herein, may be implemented by one or more components of a memory system **110** (e.g., a memory system controller **155**, a local memory controller **165**) or vice versa. In various examples, the host system **105** or an external memory controller **120** may be referred to as a host.

[0019] A processor **125** may be operable to provide functionality (e.g., control functionality) for the system **100** or the host system **105**. A processor **125** may be a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof. In some examples, a processor **125** may be an example of a central processing unit (CPU), a graphics processing unit (GPU), a general-purpose GPU (GPGPU), or an SoC, among other examples.

[0020] In some examples, the system **100** or the host system **105** may include an input component, an output component, or a combination thereof. Input components may include a sensor, a microphone, a keyboard, another processor (e.g., on a printed circuit board), an interface (e.g., a user interface, an interface between other devices), or a peripheral that interfaces with system **100** via one or more peripheral components, among other examples. Output components may include a display, audio speakers, a printing device, another processor on a printed circuit board, or a peripheral that interfaces with the system **100** via one or more peripheral components, among other examples.

[0021] The memory system **110** may be a component of the system **100** that is operable to provide physical memory locations (e.g., addresses) that may be used or referenced by the system **100**. The memory system **110** may include a memory system controller **155** and one or more memory dies **160** (e.g., memory chips) to support a capacity for data storage. The memory system **110** may be configurable to work with one or more different types of host systems **105**, and may respond to and execute commands provided by the host system **105** (e.g., via an external memory controller **120**). For example, the memory system **110** (e.g., a memory system controller **155**) may receive a write command indicating that the memory system **110** is to store data received from the host system **105**, or receive a read command indicating that the memory system **110** is to provide data stored in a memory die **160** to the host system **105**, or receive a refresh command indicating that the memory system **110** is to refresh data stored in a memory die **160**, among other types of commands and operations.

[0022] A memory system controller **155** may include components (e.g., circuitry, logic) operable to control operations of the memory system **110**. A memory system controller **155** may include hardware, firmware, or instructions that enable the memory system **110** to perform various operations, and may be operable to receive, transmit, or execute commands, data, or control information related to operations of the memory system **110**. A memory system controller **155** may be operable to communicate with one or more of an external memory controller **120**, one or more memory dies **160**, or a processor **125**. In some examples, a memory system controller **155** may control operations of the memory system **110** in cooperation with a local memory controller **165** of a memory die **160**.

[0023] Each memory die **160** may include a local memory controller **165** and a memory array **170**. A memory array **170** may be a collection of memory cells, with each memory cell being operable to store one or more bits of data. A memory die **160** may include a two-dimensional (2D) array of memory cells, or a three-dimensional (3D) array of memory cells. In some examples, a 2D memory die **160** may include a single memory array **170**. In some examples, a 3D memory die **160** may include two or more memory arrays **170**, which may be stacked or positioned beside one another (e.g., relative to a substrate).

[0024] A local memory controller **165** may include components (e.g., circuitry, logic) operable to control operations of a memory die **160**. In some examples, a local memory controller **165** may be operable to communicate (e.g., receive or transmit data or commands or both) with a memory system controller **155**. In some examples, a memory system **110** may not include a memory system controller **155**, and a local memory controller **165** or an external memory controller **120** may perform various functions described herein. As such, a local memory controller **165** may be operable to communicate with a memory system controller **155**, with other local memory controllers **165**, or directly with an external memory controller **120**, or a processor **125**, or any combination thereof. Examples of components that may be included in a memory system controller **155** or a local memory controller **165** or both may include receivers for receiving signals (e.g., from the external memory controller **120**), transmitters for transmitting signals (e.g., to the external memory controller **120**), decoders for decoding or demodulating received signals, encoders for encoding or modulating signals to be transmitted, sense components for sensing states of memory cells of a memory array **170**, write components for writing states to memory cells of a memory array **170**, or various other components operable for supporting described operations of a memory system **110**.

[0025] A host system **105** (e.g., an external memory controller **120**) and a memory system **110** (e.g., a memory system controller **155**) may communicate information (e.g., data, commands, control information, configuration information) using one or more channels **115**. Each channel **115** may be an example of a transmission medium that carries information, and each channel **115** may include one or more signal paths (e.g., a transmission medium, an electrical conductor, an electrically conductive path) between terminals associated with the components of the system **100**. For example, a channel **115** may be associated with a first terminal (e.g., including one or more pins, including one or more pads) at the host system **105** and a second terminal at the memory system **110**. A terminal may be an example of a conductive input or output point of a device of the system **100**, and a terminal may be operable to act as part of a channel **115**. In some implementations, at least the channels **115** between a host system **105** and a memory system **110** may include or be referred to as a host interface (e.g., a physical host interface). In some implementations, a host interface may include or be associated with interface circuitry (e.g., signal drivers, signal latches) at the host system **105** (e.g., at an external memory controller **120**), or at the memory system **110** (e.g., at a memory system controller **155**), or both.

[0026] In some examples, a channel **115** (e.g., associated signal paths and terminals) may be dedicated to communicating one or more types of information. For example, the channels **115** may include one or more command and address channels, one or more clock signal channels, one or more data channels, among other channels or combinations thereof. In some examples, signaling may be communicated over the channels **115** using single data rate (SDR) signaling or double data rate (DDR) signaling. In SDR signaling, one modulation symbol (e.g., signal level) of a signal may be registered for each clock cycle (e.g., on a rising or falling edge of a clock signal). In DDR signaling, two modulation symbols of a signal may be registered for each clock cycle (e.g., on both a rising edge and a falling edge of a clock signal).

[0027] In some examples, at least a portion of the system **100** may implement a stacked die architecture in which multiple semiconductor dies are physically and communicatively coupled. In some implementations, one or more semiconductor dies may include multiple instances of interface circuitry (e.g., of a memory system **110**, memory interface blocks) that are each associated with accessing a respective set of one or more memory arrays of one or more other semiconductor dies.

[0028] In accordance with examples as disclosed herein, circuitry for accessing one or more memory arrays **170** may be distributed among multiple semiconductor dies of a stack (e.g., a stack of multiple directly-coupled semiconductor dies). For example, a first die may include a logic block (e.g., a common logic block, a central logic block, logic circuitry) operable to configure a set of multiple first interface blocks (e.g., memory interface blocks, instances of first interface circuitry) of the first die. In some examples, the system may include a respective controller (e.g., a memory controller, a host interface controller, at least a portion of a memory system controller **155**, at least a portion of an external memory controller **120**, or a combination thereof) for each first interface block to support access operations (e.g., to access one or more memory arrays **170**) via the first interface block. The system may also include non-volatile storage, one or more sensors, or a combination thereof to support various operations of the system.

[0029] The multiple first interface blocks may be operable to configure operations of the multiple semiconductor dies in the stack. For example, a first interface block (e.g., of the first die) may include circuitry to perform operations, such as such as initialization operations, evaluation operations, configuration operations, access operations, scheduling operations, repair operations, refresh operations, error control operations, adverse access operations, signaling operations, or other operations based on information (e.g., signaling, instructions, parameters, configuration information) of the memory system **110**. By dividing memory access circuitry among multiple semiconductor dies in accordance with one or more of the described techniques, a system may be configured with an increased throughput of information, or a greater storage density, among other advantages, compared with other techniques for configuring a memory system.

[0030] In addition to applicability in systems as described herein, interface techniques for stacked memory architectures may be generally implemented to support artificial intelligence applications. As the use of artificial intelligence increases to support machine learning, analytics, decision making, or other related applications, electronic devices that support artificial intelligence applications and processes may be desired. For example, artificial intelligence applications may be associated with accessing relatively large quantities of data for analytical purposes and may benefit from memory systems capable of effectively and efficiently storing relatively large quantities of data or accessing stored data relatively quickly. Implementing the techniques described herein may support artificial intelligence or machine learning techniques by supporting efficient accessing of information via a relatively high quantity of closely-coupled interfaces (e.g., channels, data paths, support stacks) between a host and memory arrays of one or more semiconductor dies that are stacked over a logic die, among other benefits.

[0031] FIG. **2** illustrates an example of a system **200** (e.g., a semiconductor system, a system of coupled semiconductor dies) that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The system **200** illustrates an example of a die **205** (e.g., a semiconductor die, a host die, a processor die, a logic die) that is coupled with one or more dies **240** (e.g., dies **240**-*a*-**1** and **240**-*a*-**2**, semiconductor dies, memory dies, array dies). A die **205** or a die **240** may be formed using a respective semiconductor substrate (e.g., a substrate of crystalline semiconductor material such as silicon, germanium, silicon-germanium, gallium arsenide, or gallium nitride), or a silicon-on-insulator (SOI) substrate (e.g., silicon-on-glass (SOG), silicon-on-sapphire (SOS)), or epitaxial semiconductor materials formed on another substrate, among other examples. Although the illustrated example of a system **200** includes two dies **240**, a system **200** in accordance with the described techniques may include any quantity of one or more dies **240** coupled with a die **205**. Further, although non-limiting examples of the system **200** herein are generally described in terms of applicability to memory systems, memory sub-systems, memory devices, or a combination thereof, examples of the system **200** are not so limited. For example, aspects of the present disclosure may be applied as well to any computing system, computing sub-system, processing system, processing sub-system, component, device, structure, or other types of systems or sub-systems used for applications such as data collecting, data processing, data storage, networking, communication, power, artificial intelligence, system-on-a-chip, control, telemetry, sensing and monitoring, digital entertainment, or any combination thereof.

[0032] The system **200** illustrates an example of interface circuitry between a host and memory (e.g., via a host interface, via a physical host interface) that is implemented in (e.g., divided between) multiple semiconductor dies (e.g., a stack of directly coupled dies). For example, the die **205** may include a set of one or more interface blocks **220** (e.g., interface blocks **220**-*a*-**1** and **220**-*a*-**2**, memory interface blocks), and each die **240** may include a set of one or more interface blocks **245** and one or more memory arrays **250** (e.g., die **240**-*a*-**1** including an interface block **245**-*a*-**1** coupled with a set of one or more memory arrays **250**-*a*-**1**, die **240**-*a*-**2** including an interface block **245**-*a*-**2** coupled with a set of one or more memory arrays **250**-*a*-**2**). The memory arrays **250** may be examples of memory arrays **170**, and may include memory cells of various architectures, such as RAM, DRAM, SDRAM, SRAM, FeRAM, MRAM, RRAM, PCM, chalcogenide, NOR, or NAND memory cells, or any combination thereof.

[0033] Although the example of system **200** is illustrated with one interface block **245** included in each die **240**, a die **240** in accordance with the described techniques may include any quantity of one or more interface blocks **245**, each coupled with a respective set of one or more memory arrays **250**, and each coupled with a respective interface block **220** of a die **205**. Thus, the interface circuitry of a system **200** may include one or more interface blocks **220** of a die **205**, with each interface block **220** being coupled with (e.g., in communication with) a corresponding interface block **245** of a die **240** (e.g., external to the die **205**). In some examples, a coupled combination of an interface block **220** and an interface block **245** (e.g., coupled via a bus associated with one or more channels, such as one or more data channels, one or more control channels, one or more clock channels, or a combination thereof) may include or be referred to as a data path associated with a respective set of one or more memory arrays **250**.

[0034] In some implementations, the die **205** may include a host processor **210**. A host processor **210** may be an example of a host system **105**, or a portion thereof (e.g., a processor **125**, aspects of an external memory controller **120**, or both). The host processor **210** may be configured to perform operations that implement storage of the memory arrays **250**. For example, the host processor **210** may receive data read from the memory arrays **250**, or may transmit data to be written to the memory arrays **250**, or both (e.g., in accordance with an application or other operations of the host processor **210**). Additionally, or alternatively, a host processor **210** may be external to a die **205**, such as in another semiconductor die or other component that is coupled with (e.g., communicatively coupled with, directly coupled with, bonded with) the die **205** via one or more contacts **212**.

[0035] A host processor **210** may be configured to communicate (e.g., transmit, receive) signaling with the interface blocks **220** over one or more host interfaces **216** (e.g., a physical host interface), which may implement aspects of channels **115** described with reference to FIG. **1**. For example, the host processor **210** may be configured to transmit access signaling (e.g., control signaling, access command signaling, configuration signaling) over host interfaces **216**, which may be received by the interface blocks **220** to support access operations (e.g., read operations, write operations) on the memory arrays **250**. In some examples, a host interface **216** may include a respective set of one or more signal paths for each interface block **220**, such that the host processor **210** may communicate with each interface block **220** over the respective set of signal paths (e.g., in accordance with a selection of the respective set to perform access operations via an interface block **220** that is selected by the host processor **210**). Additionally, or alternatively, a host interface **216** may include one or more signal paths that are shared among multiple interface blocks **220** (not shown), and an interface block **220**, or a host processor **210**, or both may interpret, ignore, respond to, or inhibit response to signaling over shared signal paths of the host interface **216** based on a logical indication (e.g., an addressing indication associated with the interface block **220** or an interface enable signal, which may be provided by the host processor **210** or the corresponding interface block **220**, depending on signaling direction).

[0036] In some examples, a respective host interface **216** may be coupled between a set of one or more interface blocks **220** and a controller **215** (e.g., host interface **216-*a*-1** coupled between interface block **220-*a*-1** and controller **215-*a*-1**, host interface **216-*a*-2** coupled between interface block **220-*a*-2** and controller **215-*a*-2**). The one or more interface blocks **220** and the controller **215** may communicate (e.g., collaborate) to perform one or more operations (e.g., scheduling operations, access operations, operations initiated by a host processor **210**) associated with a memory array **250**. A controller **215** may be an example of control circuitry (e.g., memory controller circuitry, host interface control circuitry), and may be associated with implementing respective instances of one or more aspects of an external memory controller **120**, or of a memory system controller **155**, or a combination thereof for each interface block **220**. In some examples, controllers **215** may be implemented in a die **205** whether a host processor **210** is included in the die **205**, or is external to the die **205**, and the interface block **220** may communicate with the host processor **210** via one or more controllers **215**. In some other examples, controllers **215** may be implemented external to a die **205** (e.g., in another die, not shown, coupled with respective interface blocks **220** via respective terminals for each of the respective host interfaces **216**), which may be in a same die as or a different die from a die that includes a host processor **210**. In some other examples, aspects of one or more controllers **215** may be included in the host processor **210**. Although the example of system **200** is illustrated as including a controller **215** for each interface block **220**, in various examples, a controller **215** may be coupled with any quantity of one or more interface blocks **220**, and a given interface block **220** may be operable based on single controller **215**, or by one or more of a set of multiple controllers **215** (e.g., in accordance with a controller multiplexing scheme).

[0037] In some examples, a host processor **210** may determine to access an address (e.g., a logical address of a memory array **250**, a physical address of a memory array **250**, an address of an interface block **220**), and determine which controller **215** to transmit access signaling to for accessing the address (e.g., a controller **215** or interface block **220** corresponding to the address). The host processor **210** may transmit access signaling to the determined controller **215** and, in turn, the determined controller **215** may transmit access signaling to the corresponding interface block **220**. The corresponding interface block **220** may subsequently transmit access signaling to the coupled interface block **245** to access the determined address (e.g., in a corresponding memory array **250**).

[0038] A die **205** may also include a logic block **230** (e.g., a shared logic block, a central logic block, common logic circuitry), which may be configured to communicate (e.g., transmit, receive) signaling with the interface blocks **220** of the die **205**. In some cases, the logic block **230** may be configured to communicate information (e.g., commands, instructions, indications, data) with one or more interface blocks **220** to facilitate operations of the system **200**. For example, a logic block **230** may be configured to transmit configuration signaling (e.g., initialization signaling, evaluation signaling), which may be received by interface blocks **220** to support configuration of the interface blocks **220** or other aspects of operating the dies **240** (e.g., via the respective interface blocks **245**). A logic block **230** may be coupled with each interface block **220** via a respective bus **231** (e.g., bus **231-*a*-1** associated with the interface block **220-*a*-1**, bus **231-*a*-2** associated with the interface block **220-*a*-2**). In some examples, respective buses **231** may each include a respective set of one or more signal paths, such that a logic block **230** may communicate with each interface block **220** over the respective set of signal paths. Additionally, or alternatively, respective buses **231** may include one or more signal paths that are shared among multiple interface blocks **220** (not shown).

[0039] In some implementations, a logic block **230** may be configured to communicate (e.g., transmit, receive) signaling with a host processor **210** (e.g., over a bus **232**, via a contact **212** for a host processor **210** external to a die **205**) such that the logic block **230** may support an interface between the interface blocks **220** and the host processor **210**. For example, a host processor **210** may be configured to transmit initialization signaling (e.g., boot commands), or other configuration or operational signaling, which may be received by a logic block **230** to support initialization, configuration, or other operations of the interface blocks **220**. Additionally, or alternatively, in some implementations, a logic block **230** may be configured to communicate (e.g., transmit, receive) signaling with a component outside the system **200** via a bus **233** (e.g., and via a contact **234**), such that the logic block **230** may support an interface that bypasses a host processor **210**. Additionally, or alternatively, a logic block **230** may communicate with a host processor **210**, and may communicate with one or more memory arrays **250** of one or more dies **240** (e.g., to perform self-test operations for the memory arrays **250**). In some examples, such implementations may support evaluations, configurations, or other operations of the system **200**, via contacts **234** that are accessible at a physical interface of the system, during manufacturing, assembly, validation, or other operation associated with the system **200** (e.g., before coupling with a host processor **210**, without implementing a host processor **210**, for operations independent of a host processor).

[0040] Each interface block **220** may be coupled with at least a respective bus **221** of the die **205**, and a respective bus **246** of a die **240**, that are configured to communicate signaling with a corresponding interface block **245** (e.g., over one or more associated signal paths). For example, the interface block **220-*a*-1** may be coupled with the interface block **245-*a*-1** via a bus **221-*a*-1** and a bus **246-*a*-1**, and the interface block **220-*a*-2** may be coupled with the interface block **245-*a*-2** via a bus **221-*a*-2** and a bus **246-*a*-2**. In some examples, a die **240** may include a bus that bypasses operational circuitry of the die **240** (e.g., bypasses interface blocks **245** of a given die **240**), such as a bus **255**. For example, the interface block **220-*a*-2** may be coupled with the interface block **245-*a*-2** of the die **240-*a*-2** via a bus **255-*a*-1** of the die **240-*a*-1**, which may bypass interface blocks **245** of the die **240-*a*-1**. Such techniques may be extended for interconnection among more than two dies **240** (e.g., for interconnection via a respective bus **255** of multiple dies **240**).

[0041] The respective signal paths of buses **221**, **246**, and **255** may be coupled with one another, from one die to another, via various arrangements of contacts at the surfaces of interfacing dies. For example, the bus **221-*a*-1** may be coupled with the bus **246-*a*-1** via a contact **222-*a*-1** of (e.g., at a surface of) the die **205** and a contact **247-*a*-1** of the die **240-*a*-1**, the bus **221-*a*-2** may be coupled with the bus **255-*a*-1** via a contact **222-*a*-2** of the die **205** and a contact **256-*a*-1** of the die **240-*a*-1**, the bus **255-*a*-1** may be coupled with the bus **246-*a*-2** via a contact **257-*a*-1** of the die **240-*a*-1** and a contact **247-*a*-2** of the die **240-*a*-2**, and so on. Although each respective bus is illustrated with a single line, coupled via singular contacts, it is to be understood that each signal path of a given bus may be associated with respective contacts to support a separate communicative coupling via each signal path of the given bus. In some examples, a bus **255** may traverse a portion of a die **240** (e.g., in an in-plane direction, along a direction different from a thickness direction, in a waterfall arrangement), which may support an arrangement of contacts **222** along a surface of a die **205** being coupled with interface blocks **245** of different dies **240** along a stack direction (e.g., via respective contacts **256** and **257** that are non-overlapping when viewed along a thickness direction).

[0042] The interconnection of interfacing contacts may be supported by various techniques. For example, in a hybrid bonding implementation, interfacing contacts may be coupled by a fusion of conductive materials (e.g., electrically conductive materials) of the interfacing contacts (e.g., without solder or other intervening material between contacts). For example, in an assembled condition, the coupling of the die **205** with the die **240-*a*-1** may include a conductive material of the contact **222-*a*-2** being fused with a conductive material of the contact **256-*a*-1**, and the coupling of the die **240-*a*-1** with the die **240-*a*-2** may include a conductive material of the contact **257-*a*-1** being fused with a conductive material of the contact **247-*a*-2**, and so on. In some examples, such coupling may include an inoperative fusion of contacts (e.g., a non-communicative coupling, a physical coupling), such as a fusion of the contact **260-*a*-1** with the contact **256-*a*-2**, neither of which are coupled with operative circuitry of the dies **240-*a*-1** or **240-*a*-2**. In some examples, such techniques may be implemented to improve coupling strength or uniformity (e.g., implementing contacts **260**, which may not be operatively coupled with an interface block **245** or an interface block **220**), or such a coupling may be a byproduct of a repetition of components that, in various configurations, may be operative or inoperative. (e.g., where, for dies **240** with a common arrangement of contacts **256** and **257**, contacts **256-*a*-1** and **257-*a*-1** provide a communicative path between the interface block **245-*a*-2** and the interface block **220-*a*-2**, but the contacts **256-*a*-2** and **257-*a*-2** do not provide a communicative path between an interface block **245** and an interface block **220**).

[0043] In some examples, a fusion of conductive materials between dies (e.g., between contacts) may be accompanied by a fusion of other materials at one or more surfaces of the interfacing dies. For example, in an assembled condition, the coupling of the die **205** with the die **240-*a*-1** may include a dielectric material **207** (e.g., an electrically non-conductive material) of the die **205** being fused with a dielectric material **242** of the die **240-*a*-1**, and the coupling of the die **240-*a*-1** with the die **240-*a*-2** may include a dielectric material **242** of the die **240-*a*-1** being fused with a dielectric material **242** of the die **240-*a*-2**. In some examples, such dielectric materials may include an oxide, a nitride, a carbide, an oxide-nitride, an oxide-carbide, or other conversion or doping of a semiconductor material of the die **205** or dies **240**, among other materials that may support such fusion. However, coupling among dies **205** and dies **240** may be implemented in accordance with other techniques, which may implement solder, adhesives, thermal interface materials, and other intervening materials.

[0044] In some examples, dies **240** may be coupled in a stack (e.g., forming a "cube" or other arrangement of dies **240**), and the stack may subsequently be coupled with a die **205**. In some examples, a respective set of one or more dies **240** may be coupled with each die **205** of multiple dies **205** formed in a wafer (e.g., in a chip-to-wafer bonding arrangement, before cutting the wafer of dies **205**), and the dies **205**, coupled with their respective set of dies **240**, may be separated from one another (e.g., by cutting at least the wafer of dies **205**). In some other examples, a respective set of one or more dies **240** may be coupled with a respective die **205** after the die **205** is separated from a wafer of dies **205** (e.g., in a chip-to-chip bonding arrangement).

[0045] The buses **221**, **246**, and **255** may be configured to provide a configured signaling (e.g., a coordinated signaling, a logical signaling, modulated signaling, digital signaling) between an interface block **220** and a corresponding interface block **245**, which may involve various modulation or encoding techniques by a transmitting interface block (e.g., via a driver component of the transmitting interface block). In some examples, such signaling may be supported by (e.g., accompanied by) clock signaling communicated via the respective buses (e.g., in coordination with signal transmission). For example, the buses may be configured to convey one or more clock signals transmitted by the interface block **220** for reception by the interface block **245** (e.g., to trigger signal reception by a latch or other reception component of the interface block **245**, to support clocked operations of the interface block **245**). Additionally, or alternatively, the buses may be configured to convey one or more clock signals transmitted by the interface block **245** for reception by the interface block **220** (e.g., to trigger signal reception by a latch or other reception component of the interface block **220**, to support clocked operations of the interface block **220**). Such clock signals may be associated with the communication (e.g., unidirectional communication, bidirectional communication) of various signaling, such as control signaling, command signaling, data signaling, or any combination thereof. For example, the buses may include one or more signal paths for communications of a data bus (e.g., one or more data channels, a DQ bus, via a data interface of the interface blocks) in accordance with one or more corresponding clock signals (e.g., data clock signals), or one or more signal paths for communications of a control bus (e.g., a command/address (C/A) bus, via a command interface of the interface blocks) in accordance with one or more clock signals (e.g., control clock signals), or any combination thereof.

[0046] Interface blocks **220**, interface blocks **245**, and logic block **230** each may include circuitry (signaling circuitry, multiplexing circuitry, processing circuitry, controller circuitry) in various configurations (e.g., hardware configurations, logic configurations, software or instruction configurations) that support the functionality allocated to the respective block for accessing or otherwise operating a corresponding set of memory arrays **250**. For example, interface blocks **220** may include circuitry configured to perform a first subset of operations that support access of the memory arrays **250**, and interface blocks **245** may include circuitry configured to support a second subset of operations that support access of the memory arrays **250**. In some examples, the interface blocks **220** and **245** may support a functional split or distribution of functionality associated with a memory system controller **155**, a local memory controller **165**, or both across multiple dies (e.g., a die **205** and at least one die **240**). In some implementations, a logic block **230** may be configured to coordinate or configure aspects of the operations of the interface blocks **220**, of the interface blocks **245**, or both, and may support implementing one or more aspects of a memory system controller **155**. Such operations, or subsets of operations, may include operations performed in response to commands from the host processor **210**, or operations performed without commands from the host processor **210** (e.g., operations determined or initiated by an interface block **220**, operations determined or initiated by an interface block **245**, operations determined or initiated by a logic block **230**), or various combinations thereof.

[0047] In some implementations, the system **200** may include one or more instances of non-volatile storage (e.g., non-volatile storage **235** of a die **205**, non-volatile storage **270** of one or more dies **240**, or a combination thereof). In some examples, a logic block **230**, interface blocks **220**, interface blocks **245**, or a combination thereof may be configured to communicate signaling with one or more instances of non-volatile storage. For example, a logic block **230**, interface blocks **220**, or interface blocks **245** may be coupled with one or more instances of non-volatile storage via one or more buses (not shown), or respective contacts (not shown), where applicable, which may each include one or more signal paths operable to communicate signaling (e.g., command signaling, data signaling). In some examples, a logic block **230**, one or more interface blocks **220**, one or more interface blocks **245**, or a combination thereof may configure one or more operations based on information stored in one or more instances of non-volatile storage. Additionally, or alternatively, in some examples, a logic block **230**, one or more interface blocks **220**, one or more interface blocks **245**, or a combination thereof may write information (e.g., configuration information) to be stored in one or more instances of non-volatile storage.

[0048] In some implementations, the system **200** may include one or more sensors (e.g., one or more sensors **237** of a die **205**, one or more sensors **275** of one or more dies **240**, or a combination thereof). In some implementations, a logic block **230**, interface blocks **220**, interface blocks **245**, or a combination thereof may be configured to receive one or more indications based on measurements of one or more sensors of the system **200**. For example, a logic block **230**, interface blocks **220**, or interface blocks **245** may be coupled with one or more sensors via one or more buses (not shown), or respective contacts (not shown). Such sensors may include temperature sensors, current sensors, voltage sensors, counters, and other types of sensors. In some examples, a logic block **230**, one or more interface blocks **220**, one or more interface blocks **245**, or a combination thereof may configure one or more operations based on output of the one or more sensors. For example, a logic block **230** may configure one or more operations of interface blocks **220** based on signaling (e.g., indications, data) received from the one or more sensors. Additionally, or alternatively, an interface block **220** may generate access signaling for transmitting to a corresponding interface block **245** based on one or more sensors.

[0049] In some examples, circuitry of interface blocks **220**, interface blocks **245**, or a logic block **230**, or any combination thereof may include components (e.g., transistors) formed at least in part from doped portions of a substrate of the respective die. In some examples, a substrate of a die **205** may have characteristics that are different from those of a substrate of a die **240**. Additionally, or alternatively, in some examples, transistors formed from a substrate of a die **205** may have characteristics that are different from transistors formed from a substrate of a die **240** (e.g., in accordance with different transistor architectures).

[0050] In some examples, the interface blocks **220** may support a layout for one or more components within the **220**. For example, the layout may include pairing components to share an access port (e.g., a command port, a data port). Further, in some examples, the layout may support interfaces for a controller **215** (e.g., a host interface **216**) that are different from interfaces for an interface block **245** (e.g., via the buses **221**). For instance, a host interface **216** may be synchronous and have separate channels for read and write operations, while an interface via buses **221** and **246** may be asynchronous and support both read and write operations with the same channel.

[0051] A die **240** may include one or more units **265** (e.g., modules) that are separated from a semiconductor wafer having a pattern (e.g., a two-dimensional pattern) of units **265**. Although each die **240** of the system **200** is illustrated with a single unit **265** (e.g., unit **265-*a*-1** of die **240-*a*-1**, unit **265-*a*-2** of die **240-*a*-2**), a die **240** in accordance with the described techniques may include any quantity of units **265**, which may be arranged in various patterns (e.g., sets of one or more units **265** along a row direction, sets of one or more units **265** along a column direction, among other patterns). Each unit **265** may include at least the circuitry of a respective interface block **245**, along with memory array(s) **250**, a bus **251**, a bus **246**, and one or more contacts **247** corresponding to the respective interface block **245**. In some examples, where applicable, each unit **265** may also include one or more buses **255**, contacts **256**, contacts **257**, or contacts **260** (e.g., associated with a respective interface block **245** of a unit **265** of a different die **240**), which may support various degrees of stackability or modularity among or via units **265** of other dies **240**.

[0052] In some examples, the interface blocks **220** may include circuitry configured to receive first access command signaling from a host processor **210** or a controller **215** (e.g., via a host interface **216**, via one or more contacts **212** from a host processor **210** or controller **215** external to a die **205**), and to transmit second access command signaling to the respective (e.g., coupled) interface block **245** based on (e.g., in response to) the received first access command signaling. The interface blocks **245** may accordingly include circuitry configured to receive the second access command signaling from the respective interface block **220** and, in some examples, to access a respective set of one or more memory arrays **250** based on (e.g., in response to) the received second access command signaling. In various examples, the first access command signaling may include access commands that are associated with a type of operation (e.g., a read operation, a write operation, a refresh operation, a memory management operation), which may be associated with an indication of an address of the one or more memory arrays **250** (e.g., a logical address, a physical address). In some examples, the first access command signaling may include an indication of a logical address associated with the memory arrays **250**, and circuitry of an interface block **220** may be configured to generate the second access command signaling to indicate a physical address associated with the memory arrays **250** (e.g., a row address, a column address, using a logical-to-physical (L2P) table or other mapping or calculation functionality of the interface block **220**).

[0053] In some examples, to support write operations of the system **200**, circuitry of the interface blocks **220** may be configured to receive (e.g., from a host processor **210**, from a controller **215**, via a host interface **216**, via one or more contacts **212** from a host processor **210** or controller **215** external to a die **205**) first data signaling associated with the first access command signaling, and to transmit second data signaling (e.g., associated with second access command signaling) based on received first access command signaling and first data signaling. The interface blocks **245** may accordingly be configured to receive second data signaling, and to write data to one or more memory arrays **250** (e.g., in accordance with an indicated address associated with the first access command signaling) based on the received second access command signaling and second data signaling. In some examples, the interface blocks **220** may include an error control functionality (e.g., error detection circuitry, error correction circuitry, error correction code (ECC) logic, an ECC engine) that supports the interface blocks **220** generating the second data signaling based on performing an error control operation using the received first data signaling (e.g., detecting or correcting an error in the first data signaling, determining one or more parity bits to be conveyed in the second data signaling and written with the data).

[0054] In some examples, to support read operations of the system **200**, circuitry of the interface blocks **245** may be configured to read data from the memory arrays **250** based on received second access command signaling, and to transmit first data signaling based on the read data. The interface blocks **220** may accordingly be configured to receive first data signaling, and to transmit second data signaling (e.g., to a host processor **210**, to a controller **215**, via a host interface **216**, via one or more contacts **212** to a host processor **210** or controller **215** external to a die **205**) based on the received first data signaling. In some examples, the interface blocks **220** may include an error control functionality that supports the interface blocks **220** generating the second data signaling based on performing an error control operation using the received first data signaling (e.g., detecting or correcting an error in the first data signaling, which may include a calculation involving one or more parity bits received with the first data signaling).

[0055] In some examples, access command signaling that is transmitted by the interface blocks **220** to the interface blocks **245**, among other signaling, may be generated (e.g., based on access command signaling received from a host processor **210**, based on initiation signaling received from a host processor **210**, without receiving or otherwise independent from signaling from a host processor **210**) in accordance with various determination or generation techniques configured at the interface blocks **220** (e.g., based on a configuration for accessing memory arrays **250** that is modified at the interface blocks **220**). In some examples, such techniques may involve signaling or other coordination with a logic block **230**, a host processor **210**, one or more controllers **215**, one or more instances of non-volatile storage, one or more sensors, or any combination thereof. Such techniques may support the interface blocks **220** configuring aspects of the access operations performed on the memory arrays **250** by a respective interface block **245**, among other operations. For example, interface blocks **220** may include evaluation circuitry, access configuration circuitry, signaling circuitry, scheduling circuitry, repair circuitry, refresh circuitry, error control circuitry, adverse access circuitry, and other circuitry operable to configure operations associated with one or more dies (e.g., operations associated with accessing memory arrays **250** of the dies **240**). By including such components in the interface blocks **220** in accordance with one or more of the described techniques, a system may support an increased throughput of information, or a greater storage density, among other advantages, compared with other techniques for configuring a memory system.

[0056] FIG. **3** illustrates an example of an interface architecture **300** that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The interface architecture **300** illustrates an example of an interface block **245**-*b* (e.g., of a die **240**) coupled with an interface block **220**-*b* (e.g., of a die **205**). The interface block **245**-*b* may be communicatively coupled with the interface block **220**-*b* via one or more of a bus **301**, a bus **302**, a bus **303**, and a bus **304**, each of which may be examples of one or more signal paths of a bus **221** and a bus **246**, as well as a bus **255**, where applicable.

[0057] The interface block **245**-*b* includes a control interface **310** (e.g., a command interface), which may be configured to communicate signaling with the interface block **220**-*b*. For example, the control interface **310** may include circuitry (e.g., a receiver, one or more latches) configured to receive control signaling (e.g., modulated control signaling, access command signaling, configuration signaling, address signaling, such as row address signaling or column address signaling) over the bus **301**-*a*. The control interface **310** also may include circuitry configured to receive clock signaling (e.g., clock signaling associated with the control interface **310**, clock signaling having one or more phases, such as true and complement phases, dk_t/c signaling from the interface block **220**-*b*) over the bus **302**-*a*, which the control interface **310** may use for receiving the control signaling of the bus **301**-*a* (e.g., for triggering the one or more latches). The control interface **310** may transmit (e.g., forward) the control signaling over a bus **311**, and may transmit the clock signaling over a bus **312** (e.g., for timing of other operations of the interface block **245**-*b*), each of which may be received by an interface controller **320**.

[0058] The interface block **245**-*b* also includes two data interfaces **330** (e.g., data interfaces **330**-*a*-**1** and **330**-*a*-**2**), which also may be configured to communicate signaling with the interface block **220**-*b*. Each data interface **330** may include corresponding buses and circuitry, the operation of which may be associated with (e.g., controlled by, coordinated with, operated based on) control signaling via the control interface **310**. Although the example of interface block **245**-*b* includes two such data interfaces **330** associated with the control interface **310** (e.g., in a “channel pair” arrangement), the described techniques for an interface block **245** may include any quantity of one or more data interfaces **330**, and associated buses and circuitry, for a given control interface **310** of the interface block **245**. Each data interface **330** may be associated with respective data path circuitry, which may include respective first-in-first-out (FIFO) and serialization/deserialization (SERDES) circuitry (e.g., FIFO/SERDES **340**), respective write/sense circuitry **350**, respective synchronization and sequencing circuitry (e.g., sync/seq logic **360**), and respective timing circuitry **370**, along with interconnecting signal paths (e.g., one or more buses). However, in some other examples, data path circuitry may be arranged in a different manner, or may include different circuitry components, which may include circuitry that is dedicated to respective data paths, or shared among data paths, or various combinations thereof. Each data interface **330** also may be associated with a respective set of one or more memory arrays **250**. In some examples, each memory array **250** may be understood to include respective addressing circuitry such as bank logic or decoders (e.g., a row decoder, a column decoder), among other array circuitry. However, in some other examples, at least a portion of such circuitry may be included in the interface block **245**.

[0059] Each data interface **330** may include circuitry (e.g., one or more latches, one or more drivers) configured to communicate (e.g., receive, transmit) data signaling (e.g., modulated data signaling, DQ signaling) over a respective bus **303**. Each data interface **330** also may include circuitry to communicate clock signaling over a respective bus **304**, which may support clock signal reception by the data interface **330** (e.g., first clock signaling associated with the data interface **330**, clock signaling having one or more phases, such as true and complement phases, DQS_t/c signaling from the interface block **220**-*b*, clock signaling associated with data reception or write operations), or clock signal transmission by the data interface **330** (e.g., second clock signaling associated with the data interface **330**, RDQS_t/c signaling to the interface block **220**-*b*, clock signaling associated with data transmission or read operations), or both. Each data interface **330** may transmit clock signaling (e.g., received clock signaling, DQS_t/c signaling) over a respective bus **332** (e.g., for timing of other operations of the interface block **245**-*b*).

[0060] The interface controller **320** may support various control or configuration functionality of the interface block **245**-*b* for accessing or otherwise managing operations of the coupled memory arrays **250**. For example, the interface controller **320** may support access command coordination or configuration, latency or timing compensation, access command buffering (e.g., in accordance with a first-in-first-out FIFO or other organizational scheme), mode registers or logic for configuration settings, or test functionality, among other functions or combinations thereof. For each data path of the interface block **245** (e.g., associated with a respective data interface **330**), the interface controller **320** may be configured to transmit signaling to the respective memory arrays **250** over a bus **321** (e.g., address signaling, such as a row address or row activation signaling), to transmit signaling to the respective timing circuitry **370** over a bus **322** (e.g., timing signaling, which may be based on clock signaling received via the bus **312**, configuration signaling), and to transmit signaling to the respective sync/seq logic **360** over a bus **323** (e.g., timing signaling, which may be based on clock signaling received via the bus **312**, configuration signaling).

[0061] For each data path, the respective timing circuitry **370** may support timing of various operations (e.g., activations, coupling operations, signal latching, signal driving) relative to timing signaling received over a bus **322**. For example, timing circuitry **370** may include a timing chain (e.g., a global column timing chain) configured to generate one or more clock signals or other initiation signals for controlling operations of the respective data path, and such signaling may include transitions (e.g., rising edge transitions, falling edge transitions, on/off transitions) that are offset from, at a different rate than, or otherwise different from transitions of signaling over the bus **322** to support a given operation or combination of operations. For example, timing circuitry **370** may be configured to transmit signaling to the respective memory arrays **250** over a bus **371** (e.g., column selection signaling, column address signaling), to transmit signaling to the respective write/sense circuitry **350** over a bus **372** (e.g., latch or driver timing signaling), and to transmit signaling to the respective sync/seq logic over a bus **373** (e.g., timing signaling).

[0062] For each data path, the respective FIFO/SERDES **340** may be configured to convert between data signaling of a first bus width (e.g., a relatively wide bus width, associated with a bus **341** (e.g., a data read/write (DRW) bus), having a relatively larger quantity of signal paths) and a second bus width (e.g., a relatively narrow bus width, associated with a bus **331**, having a relatively smaller quantity of signal paths). In some examples, such a conversion may be accompanied by changing a rate of signaling between the bus **341** and the bus **331** (e.g., to maintain a given throughput). For example, a FIFO/SERDES **340** may support a conversion between the bus **341** having a bus width of 288 signal paths (e.g., for signaling Dat[287:0]) and the bus **331** having a bus width of 72 signal paths (e.g., for signaling DQ[71:0]), in which case a rate of signaling over the bus **331** may be four times as fast as a rate of signaling over the bus **341**. In various examples, the FIFO/SERDES may receive data signaling over the bus **331** and transmit data signaling over the bus **341** (e.g., to support a write operation), or may receive data signaling over the bus **341** and transmit data signaling over the bus **331** (e.g., to support a read operation). In some examples (e.g., to support a read operation), the FIFO/SERDES **340** may be configured to transmit clock signaling (e.g., RDQS_t/c signaling) to the data interface **330**, which may be forwarded to the interface block **220**-*b* (e.g., over a bus **304**, for reception of data signaling by the interface block **220**-*b* received over a bus **331**).

[0063] The timing or other synchronization of operations performed by the FIFO/SERDES **340** may be supported by one or more clock signals, among other signaling, received from the respective sync/seq logic **360** (e.g., over a bus **361**). For example, the sync/seq logic **360** may generate or otherwise coordinate clock signaling to support the different rates of signaling of the bus **331** and the bus **341** (e.g., based on clock signaling received over a bus **332** and a bus **373**). Additionally, or alternatively, the FIFO/SERDES **340** may operate in a direction (e.g., for data transmission to a data interface **330**, for data reception from a data interface **330**) or other mode based on configuration signaling received from the sync/seq logic **360**.

[0064] For each data path, the respective write/sense circuitry **350** may be configured to support the accessing (e.g., data signaling, write signaling, read signaling) of the respective set of one or more memory arrays **250**. For example, the write/sense circuitry **350** may be coupled with the memory arrays **250** over a bus **351** (e.g., a global input/output (GIO) bus), which may include respective signal paths associated with each memory array **250**, or may include signal paths that are shared for all of the memory arrays **250** of the set, in which case the memory array circuitry may include multiplexing circuitry operable to couple the bus **351** with a selected one of the memory arrays **250**. In some examples, a bus **351** may include a same quantity of signal paths as a bus **341** (e.g., for signaling GIO[287:0]). In some examples, a bus **351** may include a same quantity of signal paths as a quantity of columns in each memory array **250**. In some other examples, the memory arrays **250** may include a quantity of columns that is an integer multiple of the quantity of signal paths of a bus **351**, in which case the memory array circuitry (e.g., each memory array **250**) may include decoding circuitry operable to couple a subset of columns of memory cells, or associated circuitry, with the bus **351**.

[0065] To support write operations, the write/sense circuitry **350** may be configured to drive signaling (e.g., over the bus **351**) that is operable to write one or more logic states to memory cells of the memory arrays **250** (e.g., based on data received over a bus **341**, based on timing signaling received over a bus **371**, based on data signaling received over a bus **303** and on control signaling received over a bus **301**-*a*). In some examples, such signaling may be transmitted to supporting circuitry of or otherwise associated with the memory arrays **250** (e.g., as an output of signals corresponding to logic states to be written), such as sense amplifier circuitry, voltage sources, current sources, or other driver circuitry operable to apply a bias across a storage element of the memory cells (e.g., across a capacitor, across a ferroelectric capacitor), or apply a charge, a current, or other signaling to a storage element of the memory cells (e.g., to apply a current to a chalcogenide or other configurable memory material, to apply a charge to a gate of a NAND memory cell), among other examples.

[0066] To support read operations, the write/sense circuitry **350** may be configured to receive signaling (e.g., over the bus **351**) that the write/sense circuitry **350** may further amplify for communication through the interface block **245**-*b*. For example, the write/sense circuitry **350** may be configured to receive signaling corresponding to logic states read from the memory arrays **250**, but at a relatively low driver strength (e.g., relatively ‘analog’ signaling, which may be associated with a relatively low drive strength of sense amplifiers of the memory arrays **250**). The write/sense circuitry **350** may thus include further sense amplification (e.g., a data sense amplifier (DSA) between each signal path of the bus **351** and a respective signal path of the bus **341**), which each may have a relatively high drive strength (e.g., for driving relatively ‘digital’ signaling over the bus **341**).

[0067] The features of the interface architecture **300** may be duplicated in various quantities and arrangements to support a semiconductor system having multiple dies, such as various examples of a system **200**. In an example implementation, each die **240** may be configured with **64** instances of the interface block **245**-*b*, which may support a data signaling width of 9,216 signal paths for each die **240** (e.g., where each bus **303** of a channel pair is associated with 72 signal paths). For a system **200** having a stack of eight dies **240** coupled with a die **205**, the die **205** may thus be configured with **512** instances of the interface block **220**-*b*, thereby supporting an overall data signaling width of 73,738 signal paths for the system **200**. However, in other implementations, dies **205** and dies **240** may be configured with different quantities of interface blocks **220** and **245**, respectively, and a system **200** may be configured with different quantities of dies **240** per die **205**. By dividing memory access circuitry among multiple semiconductor dies (e.g., a die **205** and one or more dies **240**) in accordance with one or more of the described techniques, a system may be configured to support an increased throughput of information, or a greater storage density, among other advantages, compared with other techniques for configuring a memory system.

[0068] FIG. **4** shows an example of an interface architecture **400** that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The interface architecture **400** illustrates an example of an interface block **220**-*c* (e.g., of a die **205**) coupled with an interface block **245**-*c* (e.g., of a die **240**). The interface block **220**-*c* illustrates an example of components that may be included to support various configurations of operations for accessing memory arrays **250** (e.g., via the interface block **245**-*c*), among other operations of a system (e.g., of a system **200**). For example, the interface block **220**-*c* may include an evaluation component **405**, an access configuration component **410**, a signaling component **415**, a scheduling component **420**, a repair component **425**, a refresh component **430**, an error control component **435**, an adverse access component **445**, or any combination thereof.

[0069] One or more components of the interface block **220**-*c* may be communicatively coupled with one or more components outside the interface block **220**-*c*. For example, one or more components of the interface block **220**-*c* may be communicatively coupled with the interface block **245**-*c* via one or more of a bus **301**-*b*, a bus **302**-*b*, a bus **303**-*b*, and a bus **304**-*b*. Additionally, or alternatively, one or more components of the interface block **220**-*c* may be communicatively coupled with a host processor **210** or a controller **215** via a host interface **216**-*b*, or with a logic block **230** via a bus **231**-*b*, or both. Where applicable, the bus **231**-*b* may include one or more conductive lines that are dedicated to the interface block **220**-*c*, or one or more conductive lines that are common to a set of multiple interface blocks **220** that include the interface block **220**-*c*, or a combination thereof. In some examples, a logic block **230** that is coupled with the interface block **220**-*c* (e.g., via a bus **231**-*b*) may be operable based on signaling via a contact **234** (e.g., a pin), and the logic block **230** may be operable to configure one or more operations of or via the interface block **220**-*c* based on signaling communicated via the contact **234**. Additionally, or alternatively, a logic block **230** that is coupled with the interface block **220**-*c* may be coupled with a host processor **210**, and the logic block **230** may transmit signaling to the interface block **220**-*c* based on signaling received from the host processor **210**.

[0070] In some examples, the interface block **220**-*c* may be coupled with one or more instances of non-volatile storage (e.g., a non-volatile storage **235**, one or more instances of non-volatile storage **270**) via a bus **450**, or one or more sensors (e.g., one or more sensors **237**, one or more sensors **275**) via a bus **455**, or a combination thereof. In some examples, one or more components of the interface block **220**-*c* may configure operations based on information (e.g., instructions, parameters) received from non-volatile storage, or based on signaling associated with an output from the one or more sensors (e.g., a sensor measurement), or a combination thereof.

[0071] An evaluation component **405** may be configured to perform one or more functions such as a self-test (e.g., a built-in self-test (BIST)) functionality, manufacturing evaluations, writing initial conditions to the one or more instances of non-volatile storage, among other examples. For example, an evaluation component **405** may be configured to generate an access pattern (e.g., a pattern for accessing one or more memory arrays **250** via the interface block **245**-*c*) for evaluating various operations of the interface block **245**, of one or more memory arrays **250**, or a combination thereof, and transmit access command signaling, data signaling, or both in accordance with the generated access pattern. In various examples, an evaluation component **405** may be configured to initiate such an evaluation based on signaling received from a logic block **230** (e.g., via the bus **231**-*b*, based on signaling via a contact **234**), from a controller **215**, from the interface block **245**-*c*, instructions stored in non-volatile storage, signaling from the one or more sensors, or a combination thereof. Additionally, or alternatively, an evaluation component **405** may be configured to initiate such an evaluation based on conditions detected at the interface block **220**-*c* (e.g., based on an error detection, based on determining that an operating condition satisfies a threshold). In response to such an evaluation, the interface block **220**-*c* may modify a configuration for operations with the interface block **245**-*c* or one or more memory arrays **250** (e.g., an operating parameter, an error control configuration, a repair configuration), or may transmit an indication of a result of the evaluation (e.g., to a logic block **230**, to a controller **215**, to the interface block **245**-*c*, to a host processor **210**), among other responsive operations.

[0072] An access configuration component **410** may be configured to perform one or more access configuration functions such as interface training functions, adaptation functions, timing function, and initialization functions, among other examples. In various examples, an access configuration component **410** may configure parameters (e.g., trim parameters) the access configuration functions. For example, an access configuration component **410** may be configured to support an interface training functionality. An access configuration component **410** may be configured to synchronize, configure, or otherwise coordinate clock signal timing (e.g., frequency, phasing, offset) between the interface block **220**-*c* and the interface block **245**-*c*. In some examples, such training may involve the interface block **220**-*c* (e.g., the signaling component **415**) transmitting first clock signaling to the interface block **245**-*c* and receiving second clock signaling or other signaling from the interface block **245**-*c* (e.g., based on transmitting the first clock signaling). Based on such transmitted and received signaling (e.g., via the bus **302**-*b* and the bus **304**-*b*), the access configuration component **410** may modify a timing of the first clock signaling, or may transmit an indication to the interface block **245**-*c* to modify a timing of the second clock signaling, among other aspects of interface training.

[0073] In some examples, an access configuration component **410** may be configured to support a temperature adaptation or mitigation functionality. For example, the interface block **220**-*c* may receive an indication (e.g., from one or more sensors) of a temperature (e.g., an operating temperature, a temperature at which one or more components of a system **200** are operating), which may include an indication of a temperature itself (e.g., as an analog or digital input signal), or an indication of whether a temperature satisfies (e.g., is above, is below, is between) one or more thresholds, among other indications. Such an indication may be based on a temperature measured by a sensor outside a system **200**, or coupled with (e.g., assembled with) a system **200**, or embedded in a system **200** (e.g., one or more sensors **237** as part of a die **205**, one or more sensors **275** as part of a die **240**), or a combination thereof. The access configuration component **410** may configure the generation of access command signaling based on the temperature indication, which may include configuring a parameter for accessing a memory array **250** (e.g., a timing, an access rate, a reference voltage, a refresh rate) based on the temperature, selecting or avoiding a memory array **250** or portion thereof based on a temperature, inhibiting access operations based on a temperature, or implementing a temperature adjustment operation (e.g., a heating operation, a cooling operation), among other configurations or any combination thereof.

[0074] Additionally, or alternatively, an access configuration component **410** may be configured to support timing or latency control functionality. For example, an access configuration component **410** may be configured to control a timing of operations performed by the interface block **245**-*c*, which may refer to an absolute timing (e.g., relative to clock signaling from a host processor **210**, a controller **215**, or a logic block **230**), or a timing that is relative to timing of another interface block **220** (not shown), or both. For example, an access configuration component **410** may be configured to command operations via the interface block **245**-*c* with a timing that is offset (e.g., staggered) relative to operations commanded by one or more other interface blocks **220**. In some examples, such a staggering may balance or otherwise distribute power consumption (e.g., may reduce a ratio of peak power to average power), which may improve operational uniformity (e.g., voltage regulation uniformity) for components of a system **200**. In some examples, such techniques may be supported by an access configuration component **410** being operable to configure transmission of first clock signaling to the interface block **245**-*c* (e.g., in accordance with a first timing), and one or more other interface blocks **220** being configured to transmit second clock signaling to their respective interface block **245** with a timing that is offset relative to the first clock signaling.

[0075] A signaling component **415** may be configured to facilitate one or more signaling functions such as communication functions (e.g., transmitting functions, receiving functions), voltage swing conversion functions, among other examples. For example, a signaling component **415** may configure aspects of signaling for any function (e.g., procedures, operations) associated with the components of the interface block **220**-*c* via one or more buses (e.g., the bus **301**-*b*, the bus **302**-*b*, the bus **303**-*b*, the bus **304**-*b*, the host interface **216**-*b*, the bus **231**-*b*, the bus **450**, or the bus **455**). In some examples, a signaling component **415** may be configured to convert or otherwise provide signaling having different voltage swings. For example, a signaling component **415** may be configured to support the interface block **220**-*c* receiving first signaling (e.g., access signaling) according to a first voltage swing (e.g., associated with a die **205**) and transmitting second signaling according to a second voltage swing (e.g., associated with a die **240**, associated with the interface block **245**-*c*) that is different from the first voltage swing. In some examples, the first voltage swing may be smaller than the second voltage swing.

[0076] A scheduling component **420** may be configured to perform one or more scheduling functions such as command prioritization, command buffering, and others. For example, the interface block **220**-*c* may receive one or more commands (e.g., scheduling commands, commands for accessing a memory array **250**) from a controller **215** or from a logic block **230** and may prioritize the commands based on a type of command (e.g., read command, write command). In some examples, the scheduling component **420** may include a FIFO register for buffering one or more commands. Additionally, or alternatively, a scheduling component **420** may prioritize (e.g., schedule, optimize) one or more commands based on signaling received from a logic block **230**, instructions stored in non-volatile storage, an indication from one or more sensors, or a combination thereof. In some examples, the interface block **220**-*c* may transmit (e.g., to the interface block **245**-*c*) signaling (e.g., access signaling) according to the prioritized commands, the buffered commands, or both.

[0077] A repair component **425** may be configured to perform one or more repair functions such as repair functionality for accessing memory arrays **250** (e.g., row repair, column repair, through-silicon-via (TSV) repair, contact repair). For example, a repair component **425** may generate access command signaling based on a repair configuration such as a detected error associated with a physical address of a memory array **250** (e.g., a memory arrays **250** coupled with the interface block **245**-*c*), which may include a remapping of address space (e.g., physical addresses) by the repair component **425** to avoid accessing the physical address associated with the detected error. In some examples, a repair component **425** may generate access command signaling to indicate a row address that avoids a physical address of a failed row (e.g., row repair), or to indicate a column address that avoids a physical address of a failed column, among other examples. In some examples, a repair component **425** may generate access command signaling based on signaling from a logic block **230**, signaling from a controller **215**, signaling from the interface block **245**-*c*, instructions stored in non-volatile storage, signaling from the one or more sensors, or a combination thereof. A detected error may be associated with (e.g., may correspond to) failure or other inoperability of a row of memory cells, of a column of memory cells, or of a section of memory cells (e.g., a subarray) of a memory array **250**, or of a memory array **250** among a set of multiple memory arrays **250**, among other delineations of physical address space. In some examples, such an error may be detected by a repair component **425**, by an error control component **435**, by the interface block **245**-*c* and signaled to the interface block **220**-*c*, by a logic block **230** and signaled to the interface block **220**-*c*, by a controller **215** or a host processor **210** and indicated to the interface block **220**-*c*, or a combination thereof. In some examples, such an error may be detected in a manufacturing or validation operation, and an indication of the error may be stored in the system **200** in a manner accessible to the interface block **220**-*c* (e.g., in a register of or accessible to the interface block **220**-*c*, in an instance of non-volatile storage **235** or **270**).

[0078] Additionally, or alternatively, a repair component **425** may be configured for other memory management techniques, which may be included as part of a reliability, availability, and serviceability (RAS) solution supported by the interface block **220**-*c*. For example, a RAS solution configured at a repair component **425** may include supporting a channel data correction, such as a chip-kill or channel kill mechanism, which may involve grouping (e.g., ganging) channels together to support channel-level correctability. Additionally, or alternatively, a RAS solution configured at a repair component **425** may include supporting a cyclic redundancy check (CRC) functionality, which may be implemented for command/address integrity or data integrity. Additionally, or alternatively, a RAS solution configured at a repair component **425** may include supporting burn-in, scrubbing, test flow, qualification, or in-field BIST functionality.

[0079] Additionally, or alternatively, a RAS solution configured at a repair component **425** may include supporting a floor sweeping functionality. Floor sweeping may include a repair component **425** being operable to map out or map around regions of a die **240** (e.g., among interface blocks **245**, among memory arrays **250**) that are otherwise unrepairable. When mapping out or mapping around a region of a memory array **250**, such regions can be as small as a single row or column within a bank or channel, or may be successively larger regions, such as array sections, multiple array sections, banks, and even entire channels. In some implementations, a die **240** may be overprovisioned to allow for remapping so that elements in a dedicated region can be used for remapping within a channel or across channels to statically or dynamically replace failing regions of varying size. In some examples, such techniques may support improvements to overall product yield (e.g., yield of dies **240**, yield of systems **200**), even if such techniques may reduce an amount of overall memory capacity

[0080] A refresh component **430** may be configured to support refresh control functionality. For example, a refresh component **430** may be configured to generate an access pattern (e.g., a pattern of addresses) for performing refresh operations, and generate access command signaling based on the access pattern generated by the refresh component **430**. Such an access pattern generation may include determining a rate for refreshing memory cells of the memory arrays **250**, or addresses for refreshing, among other refresh parameters, which may be applied to each memory array **250** coupled with the interface block **245**-*c*, or specific to a certain one or more of such memory arrays **250**. In various examples, a refresh component **430** may be configured to determine a refresh rate, or other refresh parameter, based on signaling from a logic block **230**, signaling from a controller **215**, signaling from the interface block **245**-*c*, signaling from one or more sensors (e.g., temperature sensors), signaling from a host processor **210**, based on instructions stored in one or more instances of non-volatile storage, based on an evaluation of conditions by the refresh component **430** (e.g., based on an evaluation of access patterns), or various combinations thereof (e.g., based on an indicated access pattern, based on an indicated operating condition such as temperature or voltage).

[0081] An error control component **435** may be configured to perform one or more error control functions (e.g., error detection functions, error correction functions, error correcting code (ECC) operation), scrubbing operations, among other examples. For example, an error control component **435** may use parity information to correct errors detected in data received from (e.g., read from) the memory arrays **250** via the interface block **245**-*c*, or from one or more instances of non-volatile storage (e.g., non-volatile storage **235**, non-volatile storage **270**). Additionally, or alternatively, an error control component **435** may correct errors detected in data received via one or more buses (e.g., the bus **301**-*b*, the bus **303**-*b*, the host interface **216**-*b*, or the bus **231**-*b*) associated with the interface block **220**-*c*. In some examples, an error control component **435** may perform at least a portion of one or more error control operations based on signaling between the interface block **220**-*c* and a logic block **230** (e.g., via the bus **231**-*b*).

[0082] An adverse access component **445** may be configured to perform one or more adverse access mitigation functions. For example, an adverse access component **445** may be configured to support mitigation of row hammer or other adverse access patterns. An adverse access component **445** may be configured to determine (e.g., based on access command signaling received from the interface block **245**-*c*, from a host processor **210**, from a logic block **230**, from a controller **215**, or a combination thereof) that a rate of accessing one or more physical addresses of the memory arrays **250** (e.g., memory arrays **250** coupled with the interface block **245**-*c*) satisfies a threshold. Such an evaluation may include determining whether a particular row of memory cells, or other portion of a memory array **250**, is being accessed with a rate that meets or exceeds a threshold. An adverse access component **445** may be configured to generate access command signaling based on such an evaluation, which may include reducing a rate of accessing a particular physical address (e.g., one or more rows of memory cells), or inhibiting access to a particular physical address (e.g., for a configured duration), among other access modifications by the adverse access component **445** relative to received access command signaling (e.g., from a host processor **210**, from a logic block **230**, from a controller **215**). Additionally, or alternatively, an adverse access component **445** may be configured to generate refresh operation signaling based on a rate of accessing one or more physical addresses satisfying a threshold (e.g., commanding refresh operations to rows in proximity of one or more hammered rows or regions that are otherwise accessed relatively frequently), where such refresh operation signaling may involve changing a rate or time of refresh, or performing refresh operations at different rates or times among memory arrays **250** or portions thereof that are coupled with the interface block **220**-*c*, or performing refresh operations at different rates or times than another interface block **220** (not shown), among other examples. In various examples, an adverse access component **445** may be operable based on information stored in one or more instances of non-volatile storage, or based on an output of one or more sensors, or a combination thereof.

[0083] FIG. **5** shows a block diagram **500** of a memory system **520** that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The memory system **520** may be an example of aspects of a system or a memory system as described with reference to FIGS. **1** through **4**. The memory system **520**, or various components thereof, may be an example of means for performing various aspects of interface techniques for stacked memory architectures as described herein. For example, the memory system **520** may include a logic signaling component **525**, an operation component **530**, an interface signaling component **535**, an access component **540**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0084] The logic signaling component **525** may be configured as or otherwise support a means for transmitting, by logic circuitry (e.g., a logic block **230**) of a first semiconductor die (e.g., a die **205**) in communication with a plurality of first interfaces (e.g., interface blocks **220**) of the first semiconductor die, first signaling to the plurality of first interfaces. The operation component **530** may be configured as or otherwise support a means for configuring, at each of the plurality of first interfaces, one or more operations associated with the plurality of first interfaces based on the plurality of first interfaces receiving the first signaling. The interface signaling component **535** may be configured as or otherwise support a means for transmitting, based on configuring the one or more operations, respective second signaling from at least one of the plurality of first interfaces to at least one of a plurality of second interfaces (e.g., interface blocks **245**) in communication with the plurality of first interfaces, the plurality of second interfaces included in one or more second semiconductor dies (e.g., one or more dies **240**) that are coupled with the first semiconductor die. The access component **540** may be configured as or otherwise support a means for accessing, by the at least one of the plurality of second interfaces, one or more memory arrays (e.g., memory arrays **250**) of the one or more second semiconductor dies based on the at least one of the plurality of second interfaces receiving the respective second signaling.

[0085] In some examples, to support configuring the one or more operations, the operation component **530** may be configured as or otherwise support a means for configuring a trim parameter, a refresh parameter, a repair configuration, or an error control parameter, or a combination thereof, for access operations associated with the plurality of first interfaces based on the plurality of first interfaces receiving the first signaling.

[0086] In some examples, configuring the one or more operations may be based on instructions stored in non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof. In some examples, configuring the one or more operations may be based on an output of one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof.

[0087] In some examples, the described functionality of the memory system **520**, or various components thereof, may be supported by or may refer to at least a portion of a processor, where such a processor may include one or more processing elements (e.g., a controller, a microprocessor, a microcontroller, a digital signal processor, a state machine, discrete gate logic, discrete transistor logic, discrete hardware components, or any combination of one or more of such elements). In some examples, the described functionality of the memory system **520**, or various components thereof, may be implemented at least in part by instructions (e.g., stored in memory, non-transitory computer-readable medium) executable by such a processor.

[0088] FIG. **6** shows a block diagram **600** of a system **620** that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The system **620** may be an example of aspects of a system as described with reference to FIGS. **1** through **4**. The system **620**, or various components thereof, may be an example of means for performing various aspects of interface techniques for stacked memory architectures as described herein. For example, the system **620** may include a signaling component **625**, an interface component **630**, an array component **635**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0089] The signaling component **625** may be configured as or otherwise support a means for transmitting, from a host processor (e.g., a host processor **210**) to a controller (e.g., a controller **215**) of a plurality of controllers of a first semiconductor die (e.g., a die **205**), first access signaling based on determining to access an address associated with a memory array (e.g., a memory array **250**) of a plurality of memory arrays of one or more second semiconductor dies (e.g., one or more dies **240**) coupled with the first semiconductor die. The interface component **630** may be configured as or otherwise support a means for transmitting, from the controller to a first interface (e.g., an interface block **220**) of a plurality of first interfaces of the first semiconductor die, second access signaling based on the controller receiving the first access signaling. The array component **635** may be configured as or otherwise support a means for transmitting, from the first interface to a second interface (e.g., an interface block **245**) of a second semiconductor die of the one or more second semiconductor dies, third access signaling associated with accessing the memory array based on the first interface receiving the second access signaling.

[0090] In some examples, the described functionality of the system **620**, or various components thereof, may be supported by or may refer to at least a portion of a processor, where such a processor may include one or more processing elements (e.g., a controller, a microprocessor, a microcontroller, a digital signal processor, a state machine, discrete gate logic, discrete transistor logic, discrete hardware components, or any combination of one or more of such elements). In some examples, the described functionality of the system **620**, or various components thereof, may be implemented at least in part by instructions (e.g., stored in memory, non-transitory computer-readable medium) executable by such a processor.

[0091] FIG. **7** shows a flowchart illustrating a method **700** that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The operations of method **700** may be implemented by a memory system (e.g., a memory system **110**, a system **200**) or its components as described herein. For example, the operations of method **700** may be performed by a memory system as described with reference to FIGS. **1** through **5**. In some examples, a memory system may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally, or alternatively, the memory system may perform aspects of the described functions using special-purpose hardware.

[0092] At **705**, the method may include transmitting, by logic circuitry of a first semiconductor die in communication with a plurality of first interfaces of the first semiconductor die, first signaling to the plurality of first interfaces. For example, a memory system may include a logic block **230** of a die **205** that transmits first signaling to a plurality of interface blocks **220** (e.g., via buses **231**). In some examples, aspects of the operations of **705** may be performed by a logic signaling component **525** as described with reference to FIG. **5**.

[0093] At **710**, the method may include configuring, at each of the plurality of first interfaces, one or more operations associated with the plurality of first interfaces based on the plurality of first interfaces receiving the first signaling. For example, the memory system may include a plurality of interface blocks **220** that each configure one or more operations associated with the plurality of interface blocks **220** (e.g., as described with reference to FIG. **4**) based on first signaling from a logic block **230**. In some examples, aspects of the operations of **710** may be performed by an operation component **530** as described with reference to FIG. **5**.

[0094] At **715**, the method may include transmitting, based on configuring the one or more operations, respective second signaling from at least one of the plurality of first interfaces to at least one of a plurality of second interfaces in communication with the plurality of first interfaces, the plurality of second interfaces included in one or more second semiconductor dies that are coupled with the first semiconductor die. For example, the memory system may include a plurality of interface blocks **220** that transmit (e.g., via buses **301**, buses **303**, or both) respective second signaling to one or more interface blocks **245** of one or more dies **240** based on configuring one or more operations of the interface blocks **220**. In some examples, aspects of the operations of **715** may be performed by an interface signaling component **535** as described with reference to FIG. **5**.

[0095] At **720**, the method may include accessing, by the at least one of the plurality of second interfaces, one or more memory arrays of the one or more second semiconductor dies based on the at least one of the plurality of second interfaces receiving the respective second signaling. For example, the memory system may include an interface block **245** that accesses a memory array **250** (e.g., via a bus **251**) based on second signaling received from an interface block **220**. In some examples, aspects of the operations of **720** may be performed by an access component **540** as described with reference to FIG. **5**.

[0096] In some examples, an apparatus as described herein may perform a method or methods, such as the method **700**. The apparatus may include features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor), or any combination thereof for performing the following aspects of the present disclosure:

[0097] Aspect 1: A method, apparatus, or non-transitory computer-readable medium including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting, by logic circuitry of a first semiconductor die in communication with a plurality of first interfaces of the first semiconductor die, first signaling to the plurality of first interfaces; configuring, at each of the plurality of first interfaces, one or more operations associated with the plurality of first interfaces based on the plurality of first interfaces receiving the first signaling; transmitting, based on configuring the one or more operations, respective second signaling from at least one of the plurality of first interfaces to at least one of a plurality of second interfaces in communication with the plurality of first interfaces, the plurality of second interfaces included in one or more second semiconductor dies that are coupled with the first semiconductor die; and accessing, by the at least one of the plurality of second interfaces, one or more memory arrays of the one or more second semiconductor dies based on the at least one of the plurality of second interfaces receiving the respective second signaling.

[0098] Aspect 2: The method, apparatus, or non-transitory computer-readable medium of aspect 1, where configuring the one or more operations includes operations, features, circuitry, logic, means, or instructions, or any combination thereof for configuring a trim parameter, a refresh parameter, a repair configuration, or an error control parameter, or a combination thereof for access operations associated with the plurality of first interfaces based on the plurality of first interfaces receiving the first signaling.

[0099] Aspect 3: The method, apparatus, or non-transitory computer-readable medium of any of aspects 1 through 2, where configuring the one or more operations is based on instructions stored in non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof.

[0100] Aspect 4: The method, apparatus, or non-transitory computer-readable medium of any of aspects 1 through 3, where configuring the one or more operations is based on an output of one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof.

[0101] FIG. **8** shows a flowchart illustrating a method **800** that supports interface techniques for stacked memory architectures in accordance with examples as disclosed herein. The operations of method **800** may be implemented by a system (e.g., a system **200**) or its components as described herein. For example, the operations of method **800** may be performed by a system as described with reference to FIGS. **1** through **4** and **6**. In some examples, a system may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally, or alternatively, the system may perform aspects of the described functions using special-purpose hardware.

[0102] At **805**, the method may include transmitting, from a host processor to a controller of a plurality of controllers of a first semiconductor die, first access signaling based on determining to access an address associated with a memory array of a plurality of memory arrays of one or more second semiconductor dies coupled with the first semiconductor die. For example, the system may include a host processor **210** that determines to access an address of a memory array **250** of a die **240** and transmits first access signaling to a controller **215** of a die **205** based on the determination. In some examples, aspects of the operations of **805** may be performed by a signaling component **625** as described with reference to FIG. **6**.

[0103] At **810**, the method may include transmitting, from the controller to a first interface of a plurality of first interfaces of the first semiconductor die, second access signaling based on the controller receiving the first access signaling. For example, the system may include a controller **215** that transmits second access signaling to an interface block **220** of a die **205** (e.g., via a host interface **216**) based on first access signaling received from a host processor **210**. In some examples, aspects of the operations of **810** may be performed by an interface component **630** as described with reference to FIG. **6**.

[0104] At **815**, the method may include transmitting, from the first interface to a second interface of a second semiconductor die of the one or more second semiconductor dies, third access signaling associated with accessing the memory array based on the first interface receiving the second access signaling. For example, the system may include an interface block **220** that transmits third access signaling to an interface block **245** of a die **240** (e.g., via a bus **301**, a bus **303**, or both) based on second access signaling from a controller **215**. In some examples, aspects of the operations of **815** may be performed by an array component **635** as described with reference to FIG. **6**.

[0105] In some examples, an apparatus as described herein may perform a method or methods, such as the method **800**. The apparatus may include features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor), or any combination thereof for performing the following aspects of the present disclosure:

[0106] Aspect 5: A method, apparatus, or non-transitory computer-readable medium including operations, features, circuitry, logic, means, or instructions, or any combination thereof for transmitting, from a host processor to a controller of a plurality of controllers of a first semiconductor die, first access signaling based on determining to access an address associated with a memory array of a plurality of memory arrays of one or more second semiconductor dies coupled with the first semiconductor die; transmitting, from the controller to a first interface of a plurality of first interfaces of the first semiconductor die, second access signaling based on the controller receiving the first access signaling; and transmitting, from the first interface to a second interface of a second semiconductor die of the one or more second semiconductor dies, third access signaling associated with accessing the memory array based on the first interface receiving the second access signaling.

[0107] It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, portions from two or more of the methods may be combined.

[0108] An apparatus is described. The following provides an overview of aspects of the apparatus as described herein:

[0109] Aspect 6: An apparatus, including: a first semiconductor die, including: a plurality of first interfaces each including respective first circuitry operable to receive first access signaling and to transmit second access signaling based on the received first access signaling; and logic circuitry coupled with the plurality of first interfaces and operable to configure one or more operations of the plurality of first interfaces; and one or more second semiconductor dies coupled with the first semiconductor die, the one or more second semiconductor dies including: a plurality of memory arrays; and a plurality of second interfaces each coupled with a respective first interface and including respective second circuitry operable to receive the second access signaling from the respective first interface and to access a respective set of one or more of the plurality of memory arrays based on the received second access signaling.

[0110] Aspect 7: The apparatus of aspect 6, where the first semiconductor die further includes a pin coupled with the logic circuitry, the logic circuitry operable to configure the one or more operations of the plurality of first interfaces based on signaling received via the pin.

[0111] Aspect 8: The apparatus of aspect 7, where the logic circuitry is operable to perform, based on the signaling received via the pin, one or more self-test operations of the plurality of first interfaces that are associated with accessing the plurality of memory arrays.

[0112] Aspect 9: The apparatus of any of aspects 6 through 8, where the logic circuitry is coupled with a host processor of the first semiconductor die, the logic circuitry operable to configure the one or more operations of the plurality of first interfaces based on signaling received from the host processor.

[0113] Aspect 10: The apparatus of any of aspects 6 through 9, further including: non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, where the logic circuitry is operable to configure the one or more operations of the plurality of first interfaces based on information stored in the non-volatile storage.
[0114] Aspect 11: The apparatus of any of aspects 6 through 10, further including: non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, where each first interface is operable to generate the second access signaling based on information stored in the non-volatile storage.
[0115] Aspect 12: The apparatus of any of aspects 6 through 11, further including: one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, where the logic circuitry is operable to configure the one or more operations of the plurality of first interfaces based on a respective output of the one or more sensors.
[0116] Aspect 13: The apparatus of any of aspects 6 through 12, further including: one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, where each first interface is operable to generate the second access signaling based on a respective output of the one or more sensors.
[0117] Aspect 14: The apparatus of any of aspects 6 through 13, where each first interface of the plurality of first interfaces is configured to: receive the first access signaling in accordance with a first voltage swing; and transmit the second access signaling in accordance with a second voltage swing that is different from the first voltage swing.
[0118] Aspect 15: The apparatus of aspect 14, where the first voltage swing is smaller than the second voltage swing.
[0119] Aspect 16: The apparatus of any of aspects 6 through 15, where each first interface is operable to communicate with a host processor via one or more controllers of the first semiconductor die.
[0120] Aspect 17: The apparatus of any of aspects 6 through 16, where at least one second interface of the plurality of second interfaces is coupled with the respective first interface via one or more respective first conductor portions at a first surface of the first semiconductor die that are fused with one or more respective second conductor portions at a second surface of a second semiconductor die of the one or more second semiconductor dies.
[0121] Aspect 18: The apparatus of any of aspects 6 through 17, where the logic circuitry is coupled with each first interface via a respective conductive line, via a common conductive line, or via a combination thereof.
[0122] An apparatus is described. The following provides an overview of aspects of the apparatus as described herein:
[0123] Aspect 19: An apparatus, including: a first semiconductor die, including: a plurality of controllers operable to couple with a host processor; and a plurality of first interfaces each coupled with a controller of the plurality of controllers and including respective first circuitry configured to receive first signaling from the controller and to transmit second signaling based on the received first signaling; and one or more second semiconductor dies coupled with the first semiconductor die, the one or more second semiconductor dies including: a plurality of memory arrays; and a plurality of second interfaces each coupled with a respective first interface and including respective second circuitry operable to receive the second signaling from the respective first interface and to perform one or more operations on a respective set of one or more of the plurality of memory arrays based on the received second signaling.
[0124] Aspect 20: The apparatus of aspect 19, further including: non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, where each first interface is configured to generate the second signaling based on information stored in the non-volatile storage.

[0125] Aspect 21: The apparatus of any of aspects 19 through 20, further including: one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, where each first interface is configured to generate the second signaling based on an output of at least one of the one or more sensors.
[0126] Aspect 22: The apparatus of any of aspects 19 through 21, where each first interface is configured to: receive scheduling commands from the controller; and generate the second signaling based on scheduling access operations associated with one or more memory arrays of the plurality of memory arrays.
[0127] Aspect 23: The apparatus of aspect 22, where each first interface includes a first in first out (FIFO) register for buffering the scheduling commands, each first interface configured to generate the second signaling based on buffering the scheduling commands.
[0128] Aspect 24: The apparatus of any of aspects 19 through 23, further including: logic circuitry of the first semiconductor die that is coupled with the plurality of first interfaces, the logic circuitry operable to configure generation of the second signaling of at least one first interface based on third signaling from the logic circuitry to the at least one first interface.
[0129] Aspect 25: The apparatus of any of aspects 19 through 24, where one or more of the plurality of first interfaces is configured to generate the second signaling based on a trim parameter, a refresh parameter, a repair configuration, an error control parameter, or an access pattern parameter, or a combination thereof.
[0130] Aspect 26: The apparatus of any of aspects 19 through 25, where each first interface of the plurality of first interfaces is configured to: receive the first signaling in accordance with a first voltage swing; and transmit the second signaling in accordance with a second voltage swing different from the first voltage swing.
[0131] Aspect 27: The apparatus of aspect 26, where the first voltage swing is less than the second voltage swing.
[0132] Aspect 28: The apparatus of any of aspects 19 through 27, where the first semiconductor die includes a logic block coupled with the plurality of first interfaces, the logic block configured to: communicate third signaling with the host processor, fourth signaling with at least one second semiconductor die of the one or more second semiconductor dies, or both; and perform one or more self-test operations associated with one or more memory arrays of the plurality of memory arrays.
[0133] Aspect 29: The apparatus of any of aspects 19 through 28, where at least one second interface of the plurality of second interfaces is coupled with the respective first interface via one or more respective first conductor portions at a first surface of the first semiconductor die that are fused with one or more respective second conductor portions at a second surface of a second semiconductor die of the one or more second semiconductor dies.
[0134] An apparatus is described. The following provides an overview of aspects of the apparatus as described herein:
[0135] Aspect 30: An apparatus including circuitry configured to: transmit, from a host processor to a controller of a plurality of controllers of a first semiconductor die, first access signaling based on determining to access an address associated with a memory array of a plurality of memory arrays of one or more second semiconductor dies coupled with the first semiconductor die; transmit, from the controller to a first interface of a plurality of first interfaces of the first semiconductor die, second access signaling based on the controller receiving the first access signaling; and transmit, from the first interface to a second interface of a second semiconductor die of the one or more second semiconductor dies, third access signaling associated with accessing the memory array based on the first interface receiving the second access signaling.
[0136] It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, portions from two or more of the methods may be combined.

[0137] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, or symbols of signaling that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, the signal may represent a bus of signals, where the bus may have a variety of bit widths.
[0138] The terms “electronic communication,” “conductive contact,” “connected,” and “coupled” may refer to a relationship between components that supports the flow of signals between the components. Components are considered in electronic communication with (e.g., in conductive contact with, connected with, coupled with) one another if there is any electrical path (e.g., conductive path) between the components that can, at any time, support the flow of signals (e.g., charge, current, voltage) between the components. At any given time, a conductive path between components that are in electronic communication with each other (e.g., in conductive contact with, connected with, coupled with) may be an open circuit or a closed circuit based on the operation of the device that includes the connected components. A conductive path between connected components may be a direct conductive path between the components or the conductive path between connected components may be an indirect conductive path that may include intermediate components, such as switches, transistors, or other components. In some examples, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.
[0139] The term “coupling” (e.g., “electrically coupling”) may refer to condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components (e.g., over a conductive path) to a closed-circuit relationship between components in which signals are capable of being communicated between components (e.g., over the conductive path). When a component, such as a controller, couples other components together, the component initiates a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.
[0140] The term “isolated” refers to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other when the switch is open. When a controller isolates two components, the controller affects a change that prevents signals from flowing between the components using a conductive path that previously permitted signals to flow.
[0141] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details to provide an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.
[0142] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.
[0143] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions (e.g.,

code) on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

[0144] For example, the various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a processor, such as a DSP, an ASIC, an FPGA, discrete gate logic, discrete transistor logic, discrete hardware components, other programmable logic device, or any combination thereof designed to perform the functions described herein. A processor may be an example of a microprocessor, a controller, a microcontroller, a state machine, or any type of processor. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0145] As used herein, including in the claims, "or" as used in a list of items (for example, a list of items prefaced by a phrase such as "at least one of" or "one or more of") indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase "based on" shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as "based on condition A" may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase "based on" shall be construed in the same manner as the phrase "based at least in part on."

[0146] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable read-only memory (EEPROM), compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a computer, or a processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

[0147] The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. An apparatus, comprising: a first semiconductor die, comprising: a plurality of first interfaces each comprising respective first circuitry operable to receive first access signaling and to transmit second access signaling based on the received first access signaling; and logic circuitry coupled with the plurality of first interfaces and operable to configure one or more operations of the plurality of first interfaces; and one or more second semiconductor dies coupled with the first semiconductor die, the one or more second semiconductor dies comprising: a plurality of memory arrays; and a plurality of second interfaces each coupled with a respective first interface and comprising respective second circuitry operable to receive the second access signaling from the respective first interface and to access a respective set of one or more of the plurality of memory arrays based on the received second access signaling.
2. The apparatus of claim 1, wherein the first semiconductor die further comprises a pin coupled with the logic circuitry, the logic circuitry operable to configure the one or more operations of the plurality of first interfaces based on signaling received via the pin.
3. The apparatus of claim 2, wherein the logic circuitry is operable to perform, based on the signaling received via the pin, one or more self-test operations of the plurality of first interfaces that are associated with accessing the plurality of memory arrays.
4. The apparatus of claim 1, wherein the logic circuitry is coupled with a host processor of the first semiconductor die, the logic circuitry operable to configure the one or more operations of the plurality of first interfaces based on signaling received from the host processor.
5. The apparatus of claim 1, further comprising: non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, wherein the logic circuitry is operable to configure the one or more operations of the plurality of first interfaces based on information stored in the non-volatile storage.
6. The apparatus of claim 1, further comprising: non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, wherein each first interface is operable to generate the second access signaling based on information stored in the non-volatile storage.
7. The apparatus of claim 1, further comprising: one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, wherein the logic circuitry is operable to configure the one or more operations of the plurality of first interfaces based on a respective output of the one or more sensors.
8. The apparatus of claim 1, further comprising: one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, wherein each first interface is operable to generate the second access signaling based on a respective output of the one or more sensors.
9. The apparatus of claim 1, wherein each first interface of the plurality of first interfaces is configured to: receive the first access signaling in accordance with a first voltage swing; and transmit the second access signaling in accordance with a second voltage swing that is different from the first voltage swing.
10. The apparatus of claim 1, wherein each first interface is operable to communicate with a host processor via one or more controllers of the first semiconductor die.
11. The apparatus of claim 1, wherein at least one second interface of the plurality of second interfaces is coupled with the respective first interface via one or more respective first conductor portions at a first surface of the first semiconductor die that are fused with one or more respective second conductor portions at a second surface of a second semiconductor die of the one or more second semiconductor dies.
12. The apparatus of claim 1, wherein the logic circuitry is coupled with each first interface via a respective conductive line, via a common conductive line, or via a combination thereof.
13. A method, comprising: transmitting, by logic circuitry of a first semiconductor die in

communication with a plurality of first interfaces of the first semiconductor die, first signaling to the plurality of first interfaces; configuring, at each of the plurality of first interfaces, one or more operations associated with the plurality of first interfaces based on the plurality of first interfaces receiving the first signaling; transmitting, based on configuring the one or more operations, respective second signaling from at least one of the plurality of first interfaces to at least one of a plurality of second interfaces in communication with the plurality of first interfaces, the plurality of second interfaces included in one or more second semiconductor dies that are coupled with the first semiconductor die; and accessing, by the at least one of the plurality of second interfaces, one or more memory arrays of the one or more second semiconductor dies based on the at least one of the plurality of second interfaces receiving the respective second signaling.
14. The method of claim 13, wherein configuring the one or more operations comprises: configuring a trim parameter, a refresh parameter, a repair configuration, or an error control parameter, or a combination thereof, for access operations associated with the plurality of first interfaces based on the plurality of first interfaces receiving the first signaling.
15. The method of claim 13, wherein configuring the one or more operations is based on instructions stored in non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof.
16. The method of claim 13, wherein configuring the one or more operations comprises is based on an output of one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof.
17. An apparatus, comprising: a first semiconductor die, comprising: a plurality of controllers operable to couple with a host processor; and a plurality of first interfaces each coupled with a controller of the plurality of controllers and comprising respective first circuitry configured to receive first signaling from the controller and to transmit second signaling based on the received first signaling; and one or more second semiconductor dies coupled with the first semiconductor die, the one or more second semiconductor dies comprising: a plurality of memory arrays; and a plurality of second interfaces each coupled with a respective first interface and comprising respective second circuitry operable to receive the second signaling from the respective first interface and to perform one or more operations on a respective set of one or more of the plurality of memory arrays based on the received second signaling.
18. The apparatus of claim 17, further comprising: non-volatile storage of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, wherein each first interface is configured to generate the second signaling based on information stored in the non-volatile storage.
19. The apparatus of claim 17, further comprising: one or more sensors of the first semiconductor die, of at least one of the one or more second semiconductor dies, or a combination thereof, wherein each first interface is configured to generate the second signaling based on an output of at least one of the one or more sensors.
20. The apparatus of claim 17, wherein each first interface is configured to: receive scheduling commands from the controller coupled with the first interface; and generate the second signaling based on scheduling access operations associated with one or more memory arrays of the plurality of memory arrays.